# 263-3300-10L Data Science Lab: A Data Flywheel for Public, Sovereign LLM Platforms

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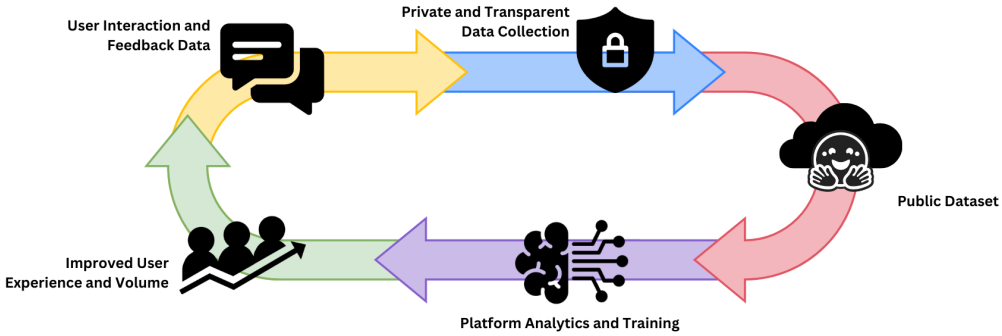



**Figure 1: Data Flywheel for Public LLM Platforms. Access: https://github.com/ethz-dslab-publicai**

## Abstract

Large Language Models (LLMs) deployed in public or sovereign contexts face a fundamental challenge between the need for high-quality user interaction data and the requirement to maintain strict standards of privacy and transparency for the citizen user-base. In this project, we design a privacy-preserving data flywheel for a public deployment of *Apertus*, enabling users to contribute chat-logs and structured feedback through an opt-in mechanism that promotes informed consent and granular control. We extend the Open WebUI platform to support contribution workflows, integrate PII stripping using Microsoft Presidio, and build a backend pipeline that submits data contributions to a public HuggingFace dataset. Our work shows that ethically governed data pipelines can yield high-utility datasets suitable for model improvement, analytics, and tooling.

## Keywords

Large Language Models, Public AI, Apertus, Data Flywheel, Privacy

## 1 Introduction

The *Apertus LLM* [2] is a Swiss sovereign large language model (LLM), hosted on the *Public AI Inference Utility* [5], and designed to align with Swiss values of transparency, privacy-preserving, and broad accessibility. In contrast to commercial LLM platforms, where data practices are often opaque and optimized to extract user value at scale, sovereign systems must demonstrate that user data is handled ethically, transparently, and in accordance with strong privacy guarantees.

High-quality user interaction data — typically consisting of chat-logs, message preferences, and annotated feedback — is essential for improving LLM reliability, safety, and alignment. Such data supports a wide range of downstream processes, such as reinforcement learning from human feedback [6], failure mode analysis, and producing training data for additional models or tools. However, existing open-source LLM interfaces lack standardized mechanisms to collect, sanitize, license, and distribute user-contributed data.

When done right, ethical data collection enables a *data flywheel*, in which user interactions create insightful data that can be used to improve the model and platform, thereby increasing user satisfaction, volume of adoption, and volume of further contributions. In this work, we research a transparent, privacy-preserving data

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Author contributions using the CRediT framework:
PA & ED: Methodology, Software, Visualization, Investigation, Writing
JT: Conceptualization, Resources, Supervision
IS: Supervision, Project Administration

flywheel for the sovereign Apertus LLM, and implement a user data collection system. Our system enables users to contribute structured chatlogs and feedback through an opt-in workflow with granular privacy controls. We demonstrate that the resulting data is suitable for analysis and training which can be integrated into model improvement and research pipelines.

## 2 Related work

Research on improving LLMs through human interaction spans several domains. Reinforcement learning from human feedback (RLHF) has become a standard technique for aligning LLM behavior with human preferences, beginning with the work of Ouyang et al. [6] and expanded to numerous commercial platforms. However, the collected datasets are typically proprietary, lacking transparency regarding collection, annotation, and privacy practices. Recent investigations into ChatGPT's data collection processes illustrate the difficulty of establishing public oversight of user-generated data in this context [1].

Open-source systems such as Open WebUI [7] support lightweight feedback mechanisms in the form of thumbs-up/thumbs-down ratings, but do not provide standardized schemas, data licenses, or privacy protocols needed to produce public, analysis-ready datasets.

Dataset creation for LLM training is typically heavily reliant on web-scraping, raising ongoing concerns around data provenance, licensing, transparency, and ethical use [3].

Our work aims to address the limitations when applying these methods to sovereign LLM platforms, by implementing privacy-preserving data collection directly within Open WebUI, combined with controlled consent and dataset publication pipelines.

## 3 Data

We collect *opt-in chatlog contributions* from users of the Public AI Inference Utility interacting with the Apertus LLM. Contributions are strictly voluntary and are initiated by users through an explicit submission workflow integrated into the user interface. Each contribution consists of a sanitized chat transcript, structured metadata, and explicit usage permissions selected by the user at submission time.

Figure 2 illustrates the structure of a data contribution. The submission schema separates data into four logical sections: (a) user identity, (b) ratings and feedback, (c) usage intent and consent, and (d) sanitized chat content review. This schema enforces consistency, and clear semantics for each data element, allowing the dataset to be directly used for analytics or training pipelines without extensive post-processing.

```
{
  "contribution_id": String,
  "contributed_at": Date,
  "attribution_mode": "anonymous" | "huggingface",
  "attribution": String,
  "ratings": {
    "accuracy": optional Number[1,5],
    "instructions": optional Number[1,5],
    "tags": optional Array[String]
  },
  "comment": optional String,
  "license_opts": {
    "model_training": Boolean,
    "qa": Boolean,
    "analytics": Boolean,
    "other": optional String,
  },
  "model": String,
  "clean_messages": [
    {
      "role": String,
      "content": String,
      "timestamp": Date,
      "feedback": optional {
        "rating": Number[1,10],
        "model_id": String,
        "comment": optional String,
        "tags": optional Array[String]
      }
    }
  ]
}
```

**Figure 2: Contribution Schema (pseudo-JSON). The standardized data structure for all contributions allows for both model training and analysis workflows.**

## 4 Methods

### 4.1 Data Collection

To collect user data on the sovereign LLM platform, the mechanism must be user-initiated and controlled, unlike automated data collection used on most commercial platforms. To this end, we extend the Open WebUI platform to add a button and form which are used to coordinate with a backend to produce a privacy-preserving data contribution.

The collection pipeline is designed with modularity as a core principle. Each component — frontend interface, backend processing, and dataset submission — can be independently customized or replaced, enabling sovereign platforms to adapt the workflow to their specific requirements while supporting code reuse across different deployments.

*4.1.1 User Initiation.* Within a conversation, below the most recent AI message, a user may click a "Contribute" button ⍐ to begin the data contribution process.

We add the button to the Open WebUI platform via an Action Function — a single-file Python script that implements an `action()` function definition — which executes whenever the associated button is pressed. This action encompasses the entirety of our backend processing. Furthermore, the action is contributed to a public, open-source repository where users may inspect the code.

*4.1.2 Chatlog Preprocessing.* Once invoked, current context of the conversation is sent to the action, including the full chatlog. At this stage, the action performs preprocessing to ensure the potential contribution is well-formed, such as checking for proper metadata and limiting the maximum number of included messages.

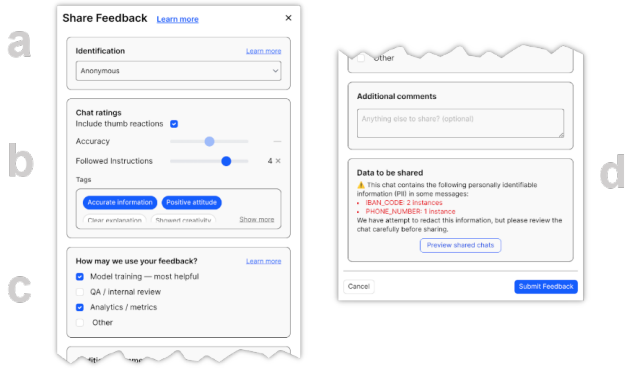



**Figure 3: Contribution Form. Data is separated into (a) User Identity, (b) Feedback, (c) Usage Intent, and (d) Chat Content, enabling both training and analytics workflows.**

If the potential contribution is invalid, it is ignored and an error is shown to the user. Otherwise, the action then cleans the chatlog of likely sensitive information. Finally, the cleaned chatlog is prepared in a JSON data format to be sent back to the frontend for user review.

*4.1.3 User Control.* After preprocessing, the action sends data back to the frontend in order to render a form to collect user preferences for which data can be shared and how their contribution should be handled.

This mechanism is enabled via *event-emitters* within Open WebUI [7]. The frontend is designed to listen for events emitted by various components in the platform, such as Action Functions. These enable coordination between the UI and backend processes, such as the display of notification toasts, or confirmation modals.

We modify the frontend Svelte code for Open WebUI's main *Chat* view to handle a new custom event of type `data_contribution`. We additionally specify that the event emission should contain additional data fields for the cleaned chatlog and counts of categories of sensitive information detected. When the action emits the data contribution event, our modified frontend code is triggered to display a custom *DataContributionDialog* UI element. This renders a Contribution Form for the user to fill out, which can be seen in Figure 3.

The contribution form is responsible for displaying information to the user about the sensitivity of their data, as well as collecting user data sharing preferences. When the form is either submitted or closed, the event is returned to the action with the form values as included data.

Once the user has submitted a contribution form, the final data contribution must be assembled according to the user's preferences. Data not selected for sharing is dropped from the action's memory, and remaining data is staged to finalize the contribution.

*4.1.4 Inclusion of Incidental Data.* In addition to chatlog data, additional metadata is collected by our system. Some fields of the contribution form, such as the user's desired attribution or additional feedback comments, are copied directly from the form to the final contribution. However, other data such as per-message ratings previously logged in the system, must be retrieved.

When a user provides a thumbs-up / down rating to a message on the Open WebUI platform, the rating is stored in a centralized SQL database. If the user specified the inclusion of message ratings when submitting their contribution form, our action connects to the SQL database and uses the schema to retrieve ratings given by the current user ID. We then perform further filtering to retain only the ratings for chat messages currently staged to contribute.

The default rating schema in Open WebUI includes a copy of the rated message, as well as other metadata such as response generation timings. In order to achieve a unified data model of contributions, we merge ratings directly into the staged chatlogs as optional metadata fields, and drop all extraneous rating metadata.

*4.1.5 Submission to HuggingFace.* Once the user confirms their contribution, the staged data is formatted into a JSON object conforming to a strict schema and submitted to a public HuggingFace Dataset repository via a pull request, which appends the contribution as a new data file. The repository URL and default API access token are configured via action settings that only administrators of the Open WebUI deployment can view and modify, ensuring that sensitive deployment details remain protected. Users who select "HuggingFace Token" as their attribution method can provide their own token, which links the contribution to their HuggingFace account and establishes verifiable provenance.

Upon successful submission, the frontend displays a status message thanking the user and providing a hyperlink to view their contribution on HuggingFace. If a user cancels the contribution at any stage, a cancellation message is displayed instead. These status notifications persist in the conversation view and are always positioned below the AI response message that prompted the contribution, providing users with clear feedback on the outcome of their action.

## 4.2 Privacy Preservation

Because the aim of the sovereign LLM platform is to direct user data into a publicly accessible dataset, it is critical that steps are taken to ensure user privacy and accordance to data usage preferences. As such, we enforce privacy preservation through multiple, complementary mechanisms designed to maximize user agency while maintaining dataset utility.

*4.2.1 In-Situ Data Processing.* During preprocessing and submission processing, it is important to note that the Action Function performs its computation on the same server where the LLM chats are processed, within the same threads. This means that even prior to data cleaning and anonymization, no data leaves the system. Only once the HuggingFace PR is created do any communications outside the system occur.

*4.2.2 PII Removal.* When a chatlog is sent to the action to initiate a contribution, we immediately apply sanitization of personally identifiable information (PII) using the well-known Microsoft Presidio library [4].

This process removes potentially sensitive text from the logs, such as names, phone numbers, credit card numbers, or social security numbers. Detection of sensitive information utilizes both regular expressions as well as semantic confidence based on nearby

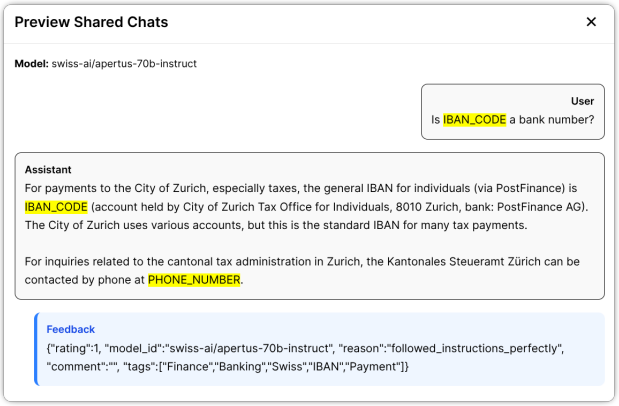



**Figure 4: Data Preview. Before submission, the user is shown all data that will be shared. Personal information is redacted automatically, and feedback is only included when specified.**

words which indicate an entity may be a specific category of information. An example of a redacted chatlog can be seen in Figure 4.

*4.2.3 User Data Control.* By leveraging a contribution form prior to submitting any data, the user is given complete control over which data elements they wish to include or exclude from their contribution.

Information such as the user's name, for attribution purposes, has a default value of "anonymous", and may be specified by the user or linked to a HuggingFace account. Other fields like additional feedback ratings and comments are entirely optional, and inclusion of thumbs-up / down ratings can be specified.

If at any time prior to submission the user decides they wish to cancel the contribution, they may do so.

*4.2.4 Data Usage Intent.* In addition to allowing the user to specify which data elements to contribute, we allow the user to specify how their data may be used. Currently, the user may select one or many of preset usage categories such as Model Training, Analytics, and Quality Assurance, or they may type their own custom intent. This approach aligns with emerging research in data licensing and decentralized consent management, where contributors maintain explicit control over the downstream uses of their data, enabling more granular and transparent governance of data-derived products.

*4.2.5 Data Transparency.* Before submission, users are presented with a comprehensive data preview displaying all content that will be shared, including the sanitized chatlog, metadata, and any selected ratings or feedback. If personally identifiable information is detected during preprocessing, a prominent warning message appears above the submit button, alerting the user and encouraging them to review their chatlog for any remaining sensitive content. Additionally, tooltips are available throughout the form to explain each field and provide links to additional documentation which clarifies the outcome of different selections, inline with the tenant of informed consent.

# 5 Results

## 5.1 Technical Feasibility

We successfully deployed a transparent, privacy-preserving data contribution pipeline on *Apertus*. The system integrates a **user-friendly UI** to initiate and review data contributions, automated **detection and redaction of personal information**, and a backend workflow that produces **consistently structured** additions to a *public dataset* for downstream use-cases. All contributions strictly respect **user-defined privacy preferences and usage permissions**. A recorded demonstration of the contribution mechanism is available via the following link:

**Figure 5: Demonstration of the final data contribution in deployment: https://youtu.be/FNy-R5yM5o8.**

## 5.2 Data Flywheel

Beyond demonstrating technical feasibility, the resulting dataset supports a range of concrete and distinct downstream applications. The modular structure of contributions—separating chat content, feedback, attribution, and usage intent—enables multiple forms of reuse without compromising privacy or consent.

Our research uncovered diverse applications, beyond RLHF, which utilize ethically-collected chatlog data to enable a flywheel. The top applications which we deemed most relevant for sovereign LLM platforms are as follows:

- *Text mining:* Aggregated, sanitized chatlogs enable analysis of recurring user intents, emerging or trending topics, and, for example, common points of confusion. Such analyses can inform policy discussions, guide public information campaigns, and help identify common user needs in AI services.
- *System and Tool development:* Developers can examine real user interactions and identify systematic failure modes, evaluate tool needs, and prioritize feature development. The structured nature of the dataset allows for targeted filtering, for example by task or need.
- *Public training data:* Contributions explicitly licensed for training can be used to train or evaluate other public-domain machine learning models. Compared to web-scraped corpora, these data offer clearer origin and explicit consent.
- *Quality assurance and benchmarking:* Contributions labeled with negative feedback or low ratings can be used to construct targeted evaluation sets, and serve as a benchmark.

These findings demonstrate that ethically collected chatlog data can serve a sustainable data flywheel. Improvements informed by contributed data have the potential to increase model and user experience quality, and user trust, which in turn encourages further participation and data contributions.

## 6 Conclusions

This work demonstrated that ethical data collection on LLM platforms is technically feasible and can sustain a privacy-preserving data flywheel. By integrating explicit consent, automated and robust sanitization, and transparent data previews into a simple and clear user interface, we show that users can retain meaningful control over their data while enabling its use.

The resulting collected data enables both traditional reinforcement learning from human feedback and novel analytics applications, such as text mining over population-scale chatlogs. More broadly, this project provides a foundation for ethical data governance in sovereign and public-sector LLM platforms, and aims to encourage further research in this direction.

## 7 Future Work

Several directions could be explored to extend this work and address its current limitations. First, future implementations could investigate Model Context Protocol (MCP)-based data collection methods in order to reduce vendor lock-in and enable contextual prompts inviting users to contribute data and/or feedback at appropriate moments during a conversation, potentially increasing participation while preserving a smooth user experience.

A second direction concerns the development of public frameworks for analyzing contributed data. This work focuses primarily on data collection, leaving data analysis to downstream users. Therefore, future work could focus on developing intuitive tools for aggregating, visualizing, and assessing the quality of the collected data. Providing these analytics tools would significantly increase the practical value of the dataset, and support an open and collaborative analytics ecosystems.

Finally, as LLM systems increasingly support multimodal interactions, future work could extend sanitization and review mechanisms to handle content such as images or audio recordings. In addition, richer consent and licensing mechanisms could be explored, allowing contributors to revise or revoke data usage permissions over time.

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